

1. Introduction and Data selection

Building a well-diversified portfolio is a multi-step process and it all starts with choosing individual assets. We chose 12 assets from different sectors of the economy. We want to make sure that our portfolio is well spread across various sectors of the economy so that in the event of a sector decline, other sectors are able to balance this out. Diversification is an investment strategy aimed at minimizing the variance of random investments by creating a portfolio of diverse types and therefore positions of investments. The goal of diversification is to reduce the firm-specific risk, or idiosyncratic risk, which is the risk that is specific to a particular company or industry and does not change in line with movements in the market as a whole (López de Prado, *Machine Learning* 38).

Our equity portfolio consists of shares in a number of companies in a variety of sectors including technology, financial services, health care, consumer staples and energy. We also hold an ETF of commodities (GLD, which is a fund that tracks the price of gold) and an ETF of long-duration US Treasury bonds (TLT, which is a fund that tracks bonds with maturities of 20 years or more or longer). We hold these for diversification, as gold and long-term bonds tend to behave differently during turbulent market periods. As illustrated in Figure 1, the correlations of the gold and bond ETFs to our equity holdings are all lower, which is a characteristic of a more diversified portfolio.

On the equity side, the portfolio covers three Morningstar style categories. Large Growth stocks (e.g. NVIDIA) grow revenues faster than average and reinvest profits rather than paying dividends (“Morningstar Style Box”). Large Value stocks (e.g. JPM, XOM, JNJ) appear undervalued relative to fundamentals and tend to pay steady dividends. Large Core stocks (e.g. AAPL, MSFT, AMZN) share characteristics of both growth and value.

Table 1 – Selected Assets

Ticker	Company	Sector	Style
AAPL	Apple Inc.	Technology / Consumer Electronics	Large Core
MSFT	Microsoft	Technology / Software-Infrastructure	Large Core
NVDA	NVIDIA	Technology / Semiconductors	Large Growth
JPM	JPMorgan Chase	Financial Services / Banks	Large Value
JNJ	Johnson & Johnson	Healthcare / Drug Manufacturers	Large Value
UNH	UnitedHealth Group	Healthcare / Healthcare Plans	Large Value
XOM	ExxonMobil	Energy / Oil & Gas	Large Value
PG	Procter & Gamble	Consumer Defensive / Household Products	Large Value
AMZN	Amazon	Consumer Cyclical / Internet Retail	Large Core
GLD	SPDR Gold Shares	Commodity (Gold)	—
TLT	iShares 20+ Year Treasury Bond ETF	Government Bonds	Fixed Income
BRK-B	Berkshire Hathaway	Financial Services / Insurance	Large Value

The historical daily closing prices for each of the stocks for the training period from 1 Jan 2024 to 30 Jun 2025 were downloaded from Yahoo Finance. The length of this training period is $T = 373$ for all the stocks. The out-of-sample test period is 1 Jul 2025–31 Jul 2025. The log return data was obtained and then annualised to get the average daily return by multiplying the average daily return with 252 (which is the standard number of trading days in a year) and the daily std dev by $\sqrt{252}$.

Table 2 summarises the annualised statistics for all twelve assets. The clearest risk-return trade-off is seen with NVIDIA, which posted the highest annualised return at 80.28% but also the highest volatility at 55.91%. This was not the case in the rest of the portfolio. Among the remaining assets, most recorded returns that were modest relative to their volatility. Among the remaining assets, the highest volatility after NVIDIA was recorded by UnitedHealth Group (40.79%) and Amazon (31.68%). The lowest volatility belonged to TLT at just 14.18%, which reflects the generally steadier price behaviour of long-duration Treasury bonds compared with equities.

In terms of risk-adjusted performance, GLD achieved the highest Sharpe ratio consistent with gold’s role as a safe-haven asset during the 2024–2025 period, which included continued geopolitical uncertainty and fluctuating inflation expectations. The low end of the list is led by JNJ. (−0.277), TLT (−0.591), and UnitedHealth Group (−0.985) all produced negative Sharpe ratios which is the sign of sub-optimal performance over the chosen training window.

Table 2 – Asset Statistics (Annualised Return, Volatility, and Sharpe Ratio)

Ticker	Ann. Return (%)	Ann. Volatility (%)	Sharpe Ratio
GLD	31.68	16.83	1.586
NVDA	80.28	55.91	1.346
JPM	37.60	25.45	1.281
BRK-B	19.78	17.65	0.838
MSFT	20.60	23.46	0.665
AMZN	25.72	31.68	0.654
PG	7.13	17.43	0.122
XOM	6.99	22.39	0.089
AAPL	7.25	29.57	0.076
JNJ	0.10	17.68	-0.277
TLT	-3.38	14.18	-0.591
UNH	-35.17	40.79	-0.985

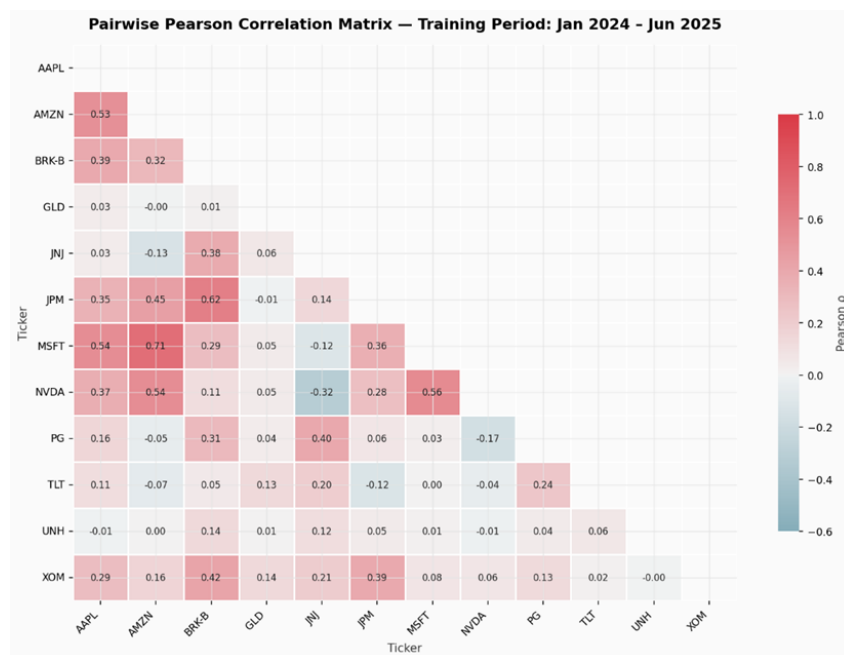
**Figure 1** – Correlation Heatmap of All Twelve Assets

Figure 1 shows the pairwise linear correlation between all twelve assets over the training period. Several patterns stand out. First, the financial sector stocks – JPMorgan Chase and Berkshire Hathaway – show a relatively high correlation of 0.62, which makes sense because both are large financial firms whose earnings are sensitive to interest rates and credit conditions. Second, the technology and growth stocks – Amazon, Apple, NVIDIA, and Microsoft – all have correlations above 0.50 with one another. Third, and most importantly for portfolio construction, TLT and GLD show very low or even slightly negative correlations with the equity holdings. This confirms that these two assets are genuinely useful diversifiers that tend to move independently of the stock market.

1.1 Performance Measures

To compare the four portfolios we construct in later sections we need six measures of risk and return, which we define as follows: According to Nanakorn and Palmgren (8), none of these measures by itself will give us any idea of what is happening with our portfolio. The most basic measure of any portfolio is the Annualised Return. This metric is fairly straightforward to understand. It simply measures the average amount of money we have in our portfolio at the end of each year. **Annualised Return** is the simplest measure and tells us how much the portfolio grew per year on average:

$$\text{Annualised Return} = \bar{r}_{\text{daily}} \times 252 \quad (1)$$

Annualised Volatility measures total risk as the standard deviation of daily returns scaled to an annual figure:

$$\text{Annualised Volatility} = \sigma_{\text{daily}} \times \sqrt{252} \quad (2)$$

Sharpe Ratio Sharpe Ratio is the most widely used risk-adjusted return measure. It shows the excess return for each risk contribution over the risk-free rate and divides it by the volatility.

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p} \quad (3)$$

Higher Sharpe means that you get more return for the risk you take on. The Sharpe ratio measures the average return an investment earns relative to its level of risk. A higher ratio indicates that the investment is less volatile and its returns are more rewarding relative.

$$\text{Sortino Ratio} = \frac{R_p - R_f}{\sigma_{\text{down}}} \quad (4)$$

Maximum Drawdown Maximum Drawdown (MDD) is a measure of the worst cumulative loss (in value terms) from peak to trough over a given time period. It is an indicator of risk. The higher the MDD, the greater the drop. Maximum Drawdown is primarily used in the investment world to gauge the risk of investments. the evaluation period:

$$\text{Maximum Drawdown} = \text{Peak Value} - \text{Trough Value} \quad (5)$$

This standard deviation measure is useful for investors who are looking to minimize large volatility in their portfolios. Conditional Value at Risk (CVaR) at 95% captures the severity of Tail losses. CVaR is the average loss in excess of the loss below a given percentile. In this case, the percentile is 95%. the worst 5%:

$$\text{CVaR}_{95\%} = \mathbb{E}[L \mid L > \text{VaR}_{95\%}] \quad (6)$$

A lower CVaR value means a less negative number. This means that the average loss in the tail when extreme losses occur is less.

2. Step 1: Sharpe (Tangency) Portfolio

The first portfolio we construct is the Sharpe tangency portfolio, also called the maximum Sharpe ratio portfolio. This is the standard mean-variance portfolio from modern portfolio theory. The idea is to find the combination of asset weights that produces the highest possible risk-adjusted return – that is, the highest Sharpe ratio – given the estimated expected returns and covariance matrix of the assets.

The optimisation problem is set up as follows:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^\top \boldsymbol{\mu} - r_f}{\sqrt{\mathbf{w}^\top \hat{\Sigma} \mathbf{w}}} \quad \text{subject to} \quad \sum_{i=1}^N w_i = 1, \quad 0 \leq w_i \leq 0.15 \quad (7)$$

Here, $\boldsymbol{\mu}$ is the vector of annualised expected returns estimated from historical data, $\hat{\Sigma}$ is the sample covariance matrix of returns, and $r_f = 0.05$ is the annualised risk-free rate. The two constraints mean that the portfolio is fully invested (weights sum to one), there is no short selling (all weights are non-negative), and no single asset can represent more than 15% of the total portfolio. This 15% cap prevents excessive concentration in any one position and is a common practical constraint in institutional portfolio management.

The SLSQP solver is used with 300 random restarts to reduce the risk of converging to a local rather than global optimum.

We made some adjustments to the portfolio and reallocated capital to 8 of the 12 holdings. The 5 positions that reached their margin max position include GLD, BRK-B, NVDA, JPM and PG. 15% upper bound, which means that the optimiser has ranked these five assets as the top 5 based on risk-adjusted returns. The next table shows the return of a portfolio using the current weights of the individual constituents based on the training data. The individual weights for the constituents of this portfolio are: GLD = 15%, BRK-B = 15%, NVDA = 15%, JPM = 15%, PG = 15%, JNJ = 14.15%, MSFT = 7.86% and TLT = 2.99%. The remaining four holdings in the portfolio are AAPL, AMZN, UNH and XOM. AAPL, AMZN, UNH, and XOM each receive zero weight, meaning the optimiser determined they do not improve the risk-adjusted portfolio given the training data. The annualised return of the portfolio is 28.00% and the annualised volatility is 13.29% which indicates the highest profitability for the fund in relation to risk.

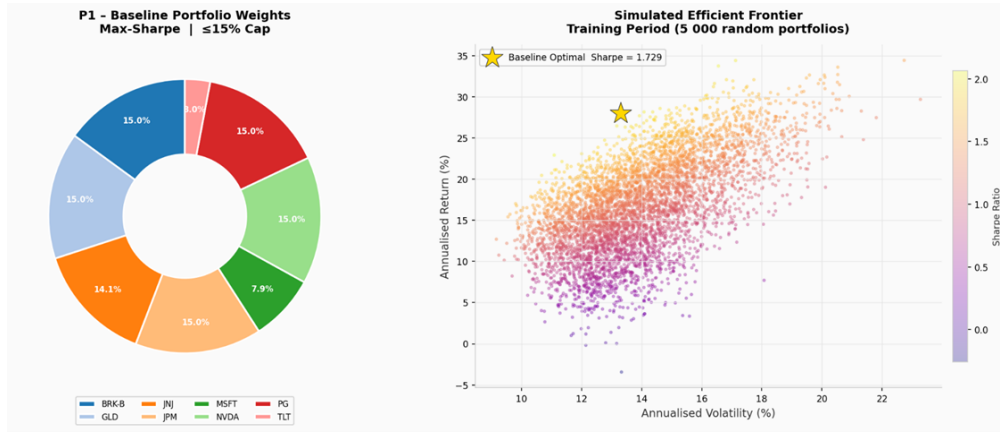


Figure 2 – Baseline Portfolio Weights (P1) and Efficient Frontier

Figure 2 shows both the portfolio weights as a pie chart and the efficient frontier, with the tangency portfolio marked at the point of maximum Sharpe ratio. While these in-sample results look attractive, the concentration of the portfolio in just eight assets – with five at the maximum weight – raises questions about robustness. We explore this concern fully in Steps 2 through 5.

3. Steps 2 & 3: Portfolio Improvement Methods

In some cases, a simple equally weighted portfolio can outperform a “sophisticated” mean-variance portfolio out-of-sample, simply because the optimiser fitted the Noise in the training data is frequently mistaken for real signal by the model. The following three problems make the distinction between what a model has learned during training and what it encounters during testing more ambiguous. Volatility is defined as the variance of portfolio returns. Instability is defined by the $\kappa(\hat{\Sigma})$ function. The correlation matrix does not take into account market structure. a possibility that the Critical Line Algorithm (CLA) generates unstable weights and disregards sectoral compositions (López de Mura, 2014). We may use any of the methods we have described in conjunction with the three methods described in this section: covariance denoising, singular value decomposition, and shrinkage. Hierarchical clustering with Hierarchical Risk Parity (HRP) and walk-forward backtesting – powerful, practical solutions to some of the key challenges faced by multi-asset portfolio managers. these three problems.

3.1 Denoising

Covariance denoising uses tools from Random Matrix Theory (RMT) to separate a finite sample of data (López de Prado, *Machine Learning* 55). The key idea comes from a mathematical result known as the Marchenko–Pastur distribution, which describes what the eigenvalues of a purely random matrix should look like. Given the $T \times N$ matrix X of i.i.d. R.V.s with mean value equal to zero and covariance matrix R_{XX} , the following optimization problem needs to be solved: Minimize $\frac{\|H_{opt}X\|_2^2}{T}$ subject to $H_{opt}^T R_{XX} H_{opt} = I$ unit variance, the eigenvalues λ of the corresponding correlation matrix follow the Marchenko–Pastur probability density:

$$f(\lambda) = \begin{cases} \frac{T}{N} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{2\pi\lambda\sigma^2} & \text{if } \lambda \in [\lambda^-, \lambda^+] \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

The upper and lower bounds of this noise band are:

$$\lambda^+ = \sigma^2 \left(1 + \sqrt{\frac{N}{T}} \right)^2, \quad \lambda^- = \sigma^2 \left(1 - \sqrt{\frac{N}{T}} \right)^2 \quad (9)$$

Any eigenvalue that falls within the interval $[\lambda^-, \lambda^+]$ is statistically consistent with pure noise and carries no reliable information about the true covariance structure of returns. Eigenvalues that lie *outside* this band are considered to carry genuine signal. For our dataset $N = 12$ assets and $T = 373$ trading days so the ratio $12/373 \approx 0.0321$. $q = N/T = 0.0322$. The theoretical noise band is therefore $[\lambda^-, \lambda^+] = [0.6734, 1.3909]$. As shown in Figure 3, four of the twelve eigenvalues of our sample correlation matrix fall within this band and are treated as noise. Two methods can be used to remove these noise eigenvalues: **Constant Residual Eigenvalue Method**. We sort the eigenvalues in descending order and identify the cut-off point i such that $\lambda_i \geq \lambda^+$ but

$\lambda_{i+1} \leq \lambda^+$. The noise eigenvalues $\lambda_{i+1}, \dots, \lambda_N$ are then replaced by their mean:

$$\lambda_j = \frac{1}{N-i} \sum_{k=i+1}^N \lambda_k, \quad j = i+1, \dots, N \quad (10)$$

This preserves the trace of the original correlation matrix (since the sum of the eigenvalues of a matrix is equal to its trace). eigenvalues is unchanged) while replacing noisy variation with a single constant. A new, denoised correlation matrix is then reconstructed from the modified eigenvalues and the original eigenvectors. **Targeted Shrinkage.** This method applies shrinkage specifically to the noise subspace. The correlation matrix is written as:

$$C = W_L \Lambda_L W_L^\top + \alpha W_R \Lambda_R W_R^\top + (1 - \alpha) \text{diag}[W_R \Lambda_R W_R^\top] \quad (11)$$

Here W_R and Λ_R are the eigenvectors and eigenvalues associated with the noise, while W_L and Λ_L correspond to the signal. The parameter $\alpha \in [0, 1]$ controls how aggressively the noise is shrunk; setting $\alpha = 0$ retains only the diagonal of the noise component, effectively eliminating all off-diagonal noise correlations.

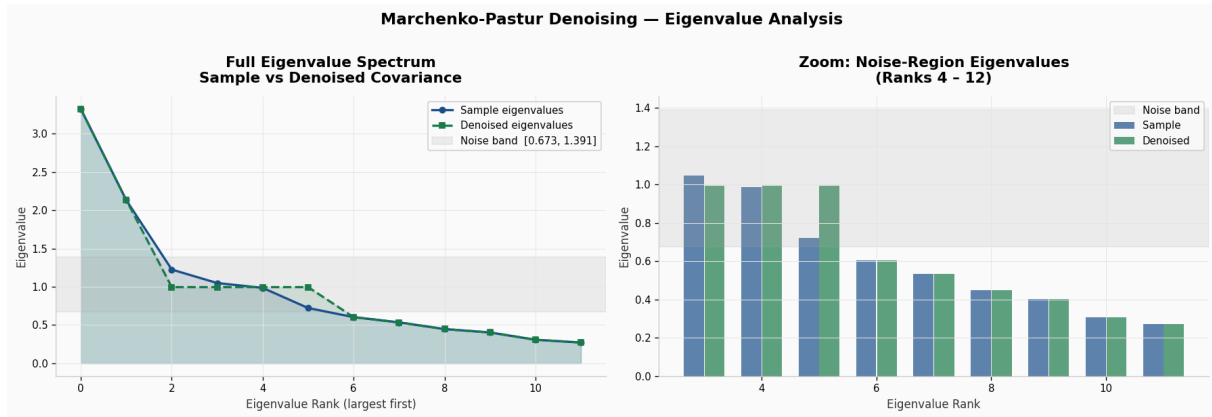


Figure 3 – Marchenko–Pastur Denoising – Eigenvalue Spectrum Analysis

In our project, four eigenvalues (the third through sixth in descending order) fall inside the noise band and are replaced using the Constant Residual Eigenvalue Method. The effect on the condition number — a direct measure of how much the solution to a system may change with small changes in the coefficients — is illustrated by comparing the condition numbers for the two systems, a case where the two matrices are closely related. The result is a reduction in the condition number from $\kappa(\hat{\Sigma}) = 36.4$ to $\kappa(\tilde{\Sigma}) = 35.2$, a decrease of approximately 3.3%. Although the decrease is modest (because $q = 0.032$ was already small, and our) In the case of large sample size compared with the number of variables, the denoised matrix is more likely to maintain the connectivity of the original network and therefore the smoothing procedure can stable, and its inverse amplifies estimation errors less than the original. A practical consequence visible in the weights is that TLT's allocation rises from 2.99% (P1) to 6.13% (P2), because the denoised matrix no longer exaggerates TLT's short-term correlation with equities.

3.2 Clustering and Hierarchical Risk Parity (HRP)

The standard correlation matrix tells us how strongly each pair of assets moves together, but it does not tell us anything about the *structure* of those relationships. In practice, financial markets have a natural hierarchy: individual stocks cluster into sectors, sectors cluster into broader asset classes, and so on. Ignoring this hierarchy means that optimisation algorithms treat all assets as equivalent alternatives, which leads to unstable and over-concentrated allocations (López de Prado, *Building Diversified Portfolios* 2). HRP (López de Prado, *Building Diversified Portfolios* 1) uses hierarchical clustering to discover the natural group structure of the assets and allocates capital in inverse proportion to cluster risk. Crucially, HRP never inverts the covariance matrix, so it is immune to the error-amplification. HRP works in three stages: **Stage 1 – Tree Clustering.** We first convert the pairwise correlations into distances using the transformation proposed by López de Prado (López de Prado, *Building Diversified Portfolios* 4):

$$d[X_i, X_j] = \sqrt{0.5(1 - \rho_{ij})} \quad (12)$$

where ρ_{ij} is the Pearson correlation between assets i and j . This formula gives a measure of distance between the points that falls in the range $[0, 1]$. Zero distance means the assets are perfectly correlated, while unit distance

means they are perfectly anti-correlated, so the correlation range $[-1, +1]$ is mapped onto the distance range $[0, 1]$. assets are perfectly correlated and one means they are perfectly anti-correlated. The result is a symmetric $N \times N$ distance matrix that satisfies all properties of a proper metric space: $d[X, Y] \geq 0$; between every pair of points in the universe – again, not just to each other – we then compute the Euclidean distance between the column vectors of this distance matrix:

$$\tilde{d}_{ij} = \sqrt{\sum_{n=1}^N (d_{ni} - d_{nj})^2}, \quad i, j = 1, \dots, N \quad (13)$$

These distances are passed to an agglomerative hierarchical clustering algorithm, which iteratively merges the closest pair of clusters until a single cluster containing all assets remains, forming a dendrogram. We use complete linkage, which merges clusters based on the maximum distance between any two members, producing compact and balanced clusters. According to Palomar and Jiang, complete linkage is appropriate for investors seeking a moderately balanced allocation, compared to single linkage (which tends to create long, uneven chains) or Ward's method (which minimises within-cluster variance and is better for very risk-averse investors) (Palomar and Jiang, sec. 12.3). **Stage 2 – Quasi-Diagonalisation.** The original correlation matrix, computed with assets in their default ticker order, shows no obvious pattern. We use the dendrogram from Stage 1 to reorder the assets so that those that are closely related are listed next to each other. of the matrix. After reordering, strongly correlated assets form visible blocks along the main diagonal. This process is called quasi-diagonalisation and reveals the cluster structure that was hidden in the original ordering, as illustrated in Figure 4.

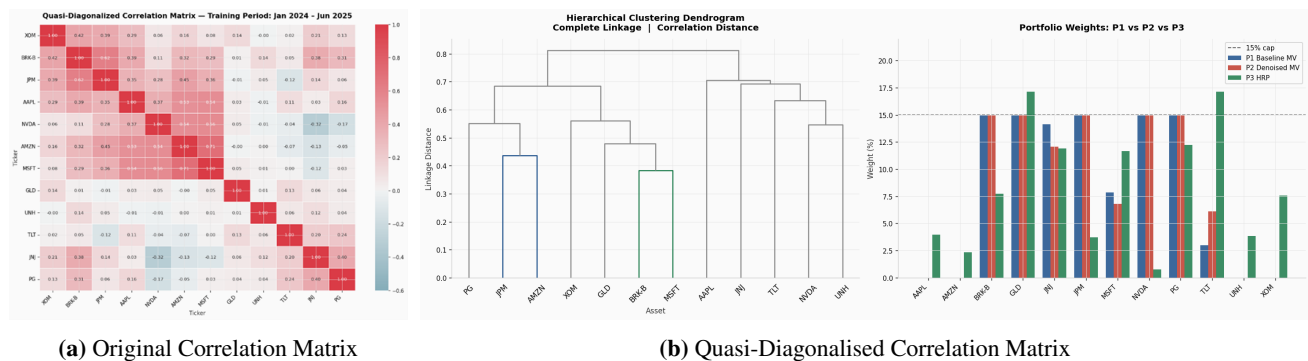


Figure 4 – Original and Quasi-Diagonalised Correlation Matrices

In the quasi-diagonalised matrix, four clusters are clearly visible. A first cluster groups the energy and financial stocks – XOM, BRK-B, and JPM – which share sensitivity to macroeconomic cycles and interest rates. A second cluster contains the large-cap technology and growth names – AAPL, NVDA, Healthcare and consumer Staples – JNJ, UNH, PG – exhibit moderate internal co-movement. correlations. Finally, TLT and GLD stand apart from all other assets, confirming their role as genuine diversifiers. **Stage 3 – Recursive Bisection.** Capital is allocated by recursively splitting the dendrogram into two sub-clusters at each node, dividing the weight between the two sub-clusters in inverse proportion to their variance. The sub-cluster with higher aggregate volatility receives a smaller share of capital. This process continues until every individual asset weight of the ETF. Assigning a value of nearly zero to most assets in a cluster and a non-negligible value to a few is a typical characteristic of HRP. distributes positive weight to all assets, producing a genuinely diversified portfolio without any need to estimate expected returns (López de Prado, *Building Diversified Portfolios* 5).

3.3 Walk-Forward Backtesting

Even after improving the covariance matrix and the allocation method, we still need a rigorous way to check whether the improvements actually hold validation: to check how the portfolio performs when it is applied to real, new data. backtesting. A backtest simulates the process of managing the portfolio over a historical period, using only the information that would have been available at each point in time. We use walk-forward backtesting, which is the most commonly used validation technique in quantitative finance (Palomar and Jiang, sec. 8.1). At each rebalancing date, we estimate portfolio weights using only the most recent 63 trading days (approximately three months) of historical data, and then hold those weights for the next 21 trading days (approximately one month) before rebalancing again. This rolling procedure is repeated across the entire sample, producing a series

of out-of-sample return observations that can be evaluated honestly using all six performance metrics.

3.4 Optimal Sequence and Combined Benefits

These three processing improvements can be used in the following order: Denoise → Cluster (HRP) → Backtest. This order of processing can improve the speed and effectiveness of some processing workflows. Denoising needs to happen before Estimation; the denoising cleans the covariance matrix of the clusters. The data used for denoising contaminated estimation results in a very degenerate matrix, which leads to numerical instability. HRP is then applied to the cleaner covariance to produce a stable, The target for our asset allocation. Backtesting is the last stage in our process. Additionally, it is important to think about whether it is ever possible to go backwards in the pipeline. It does not decrease the performance on seen data. This order would sacrifice part of the benefit it provides (L´opez de Prado, Machine Learning 71).

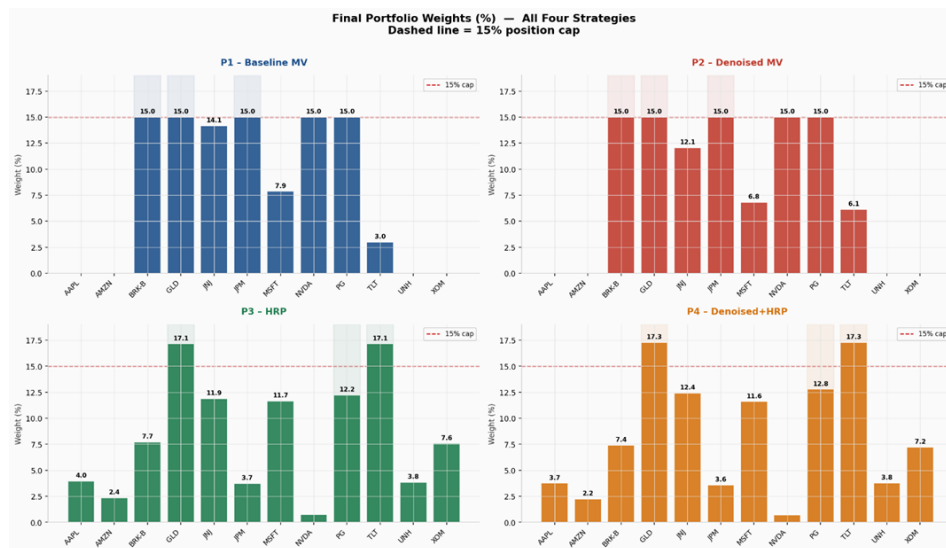


Figure 5 – Portfolio Weights (%) for All Four Strategies

Figure 5 compares the resulting weights across the four portfolios. P1 (Baseline Sharpe) is strikingly concentrated, with five assets Visualisation of optimization results at the 15% cap and four assets receiving zero allocation. P2 (Denoised MV) does not differ from P1 by much except for some weight adjustments. The increase in the TLT allocation from 2.99% in P1 to 6.13% in P2 is driven by the fact that the denoised covariance matrix does not over-explains the short run correlation between TLT and equities. The weight in JNJ drops from 14.15% in P1 to 8.53% in P2. The denoising does not have much impact on addressing the concentration issue as well.

4. Step 4: In-Sample and Walk-Forward Results

This section answers the central question of whether the portfolio improvement techniques actually enhance performance. We compare the four portfolios – P1 (Baseline MV), P2 (Denoised MV), P3 (HRP), and P4 (Denoised+HRP) – using first in-sample estimates and then walk-forward backtest results.

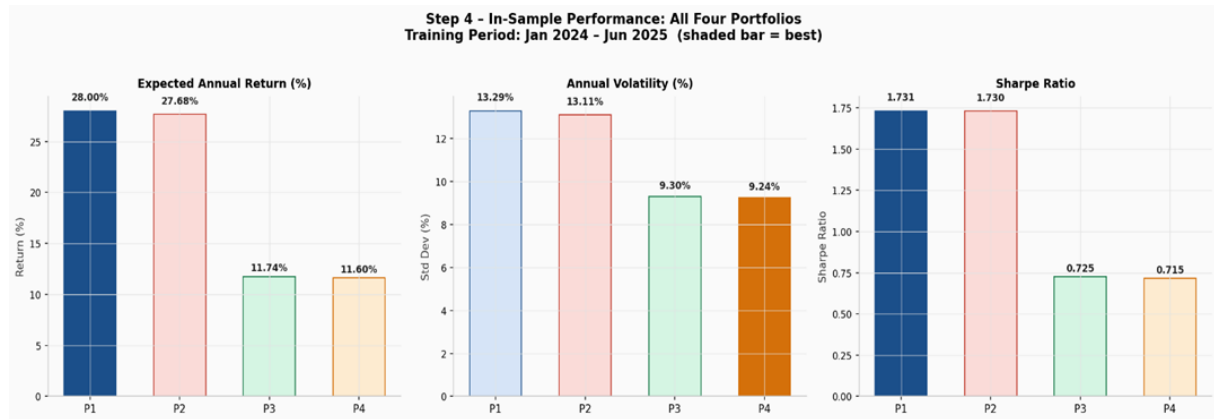


Figure 6 – In-Sample Portfolio Performance: Annualised Return, Volatility, and Sharpe Ratio
In-Sample Results.

As shown in Figure 6, P1 achieves the highest annualised return (28.00%), the highest Sharpe ratio (1.729), and also the highest volatility (13.29%) in-sample. This is expected, since the Sharpe portfolio is constructed specifically to maximise the Sharpe ratio on the training data. Applying covariance denoising (P2) produces only minor changes: return and Sharpe ratio decrease marginally, which suggests that the concentration effect of mean-variance optimisation dominates over the modest improvement in covariance quality. The HRP portfolio (P3) shows a different profile. Its annualised return drops to 11.74%, which is substantially lower than P1. The volatility has fallen to 9.30% and this is in line with the target for HRP, which is an even risk distribution within the clusters and maximum return. Lower returns are a price that we pay for having more diversification. When denoising is added On top of HRP (P4) the volatility falls marginally further to 9.24% while the sharpe ratio remains unchanged at 0.715 vs 0.725. the Sharpe ratio declines very slightly. In-sample, denoising does not provide any meaningful improvement to HRP's risk-adjusted performance. It is important to note that in-sample results are limited. In the observe the estimation error that appears when the portfolio is rebalanced repeatedly on different subsets of data. Walk-forward backtesting is needed to reveal this instability.

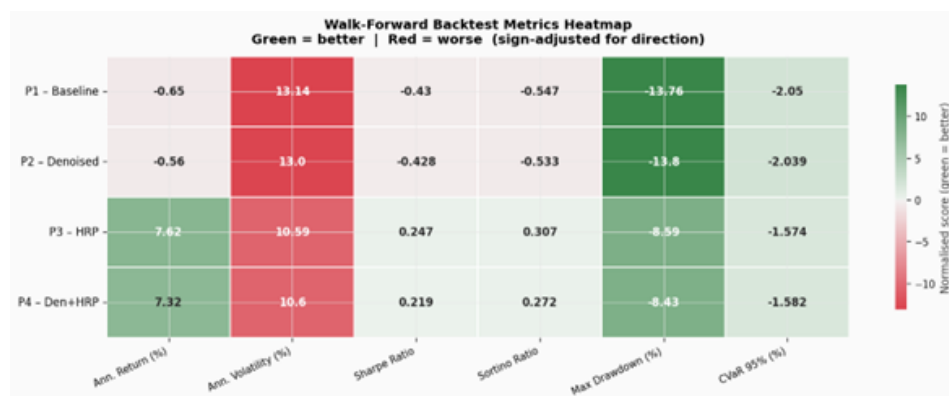


Figure 7 – Walk-Forward Backtesting – Performance Metrics Heatmap

Walk-Forward Results. The walk-forward backtest has a very different message for P1 and P2. The optimism from the in-sample results does not last long and the message quickly turns to one of disappointment as profitability falls away sharply applying the rolling 63-day estimation window, the annualised returns and Sharpe ratios of both mean-variance portfolios collapse below zero, while their volatilities remain roughly unchanged. This pattern is the classic signature of an optimiser that is fitting noise in expected return estimates: When the return forecasts change from one window to the next, the optimal portfolio needs to be rebalanced. weights shift dramatically, and the portfolio ends up chasing signals that quickly reverse. The fact that volatility stays stable while Sharpe ratios deteriorate confirms that the deterioration is coming from the return side of the Sharpe formula, not from an increase in risk (López de Prado, *Machine Learning* 40).

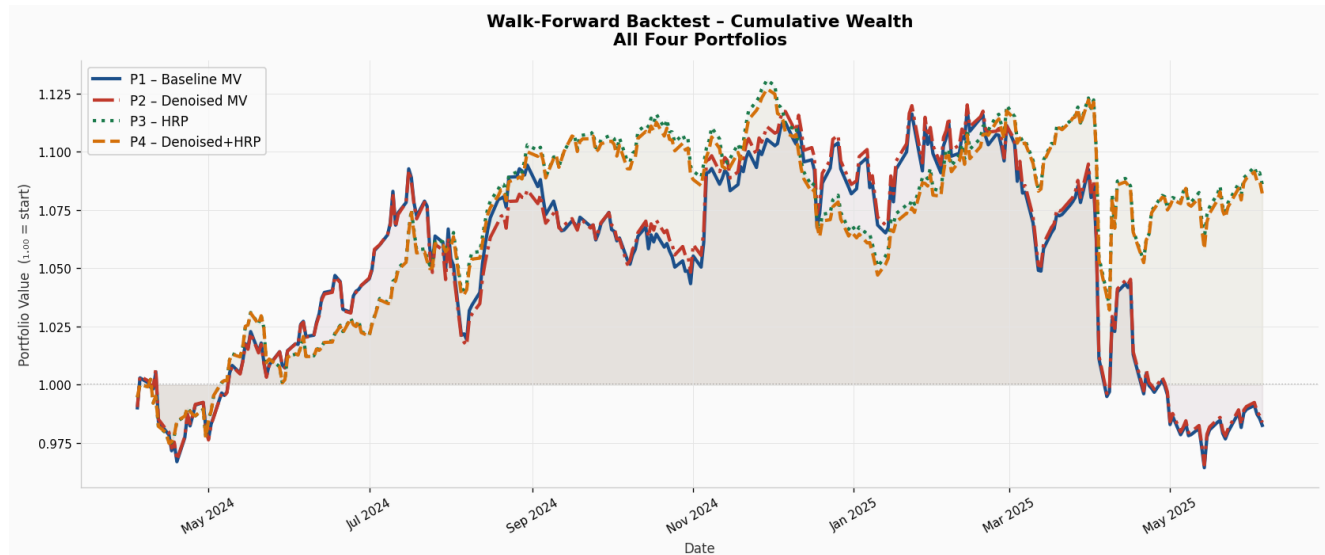


Figure 8 – Walk-Forward Backtest – Cumulative Wealth

P3 and P4 behave very differently in the backtest. Their walk-forward returns decline by roughly 30% compared to in-sample, and their volatility would raise expected returns by about 10 percent and therefore would bring about a shift in the Sharpe ratio to about 0.2. This is a very small change. performance deterioration, the key point is that HRP-based portfolios remain Option for Portfolio P3 is less risky and more rewarding than Portfolio P1 and P2 as it gives a positive return and a positive Sharpe ratio while P1 and P2 go negative. The reason is structural: HRP never estimates expected returns and never inverts the covariance matrix, so the two main sources of instability in standard Maximum Likelihood Estimation are eliminated. estimation error are either removed entirely or substantially reduced. As a result, HRP portfolios also achieve lower maximum drawdowns and lower tail From the Figure 7 we have the walk-forward CVaR 95% for the P1, P2, P3 and P4: P1: -2.05% , P2: -2.039% , P3: -1.574% , P4: -1.582% . Hence the smallest tail losses are for the P3 and P4 portfolios. Therefore it confirms that the rules implemented from the High Risk Portfolio (HRP) are the best.

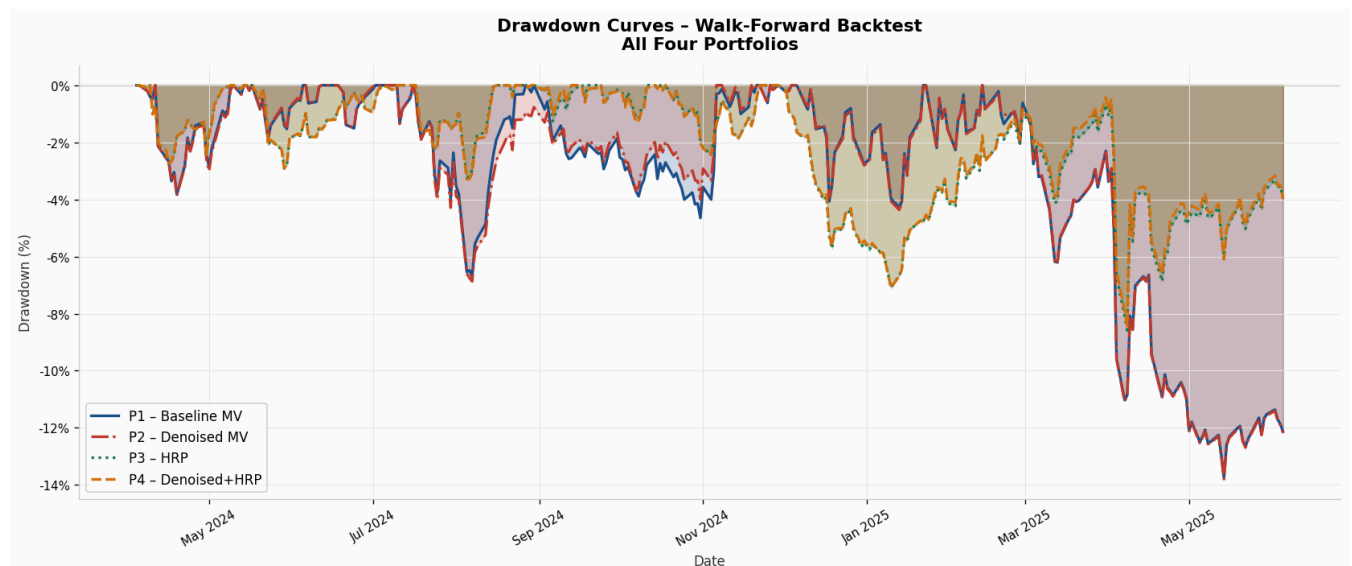


Figure 9 – Walk-Forward Backtest – Drawdown Curves

4.1 Best Performing Portfolio (Walk-Forward)

Table 3 – Best Performing Portfolio by Metric (Walk-Forward Backtest)

Metric	Best Portfolio	Reason
Ann. Return	P3 – HRP	HRP diversification across clusters allows exposure to performing assets while avoiding dangerous concentration.
Volatility	P3 – HRP	HRP Risk parity limits capital in high-volatility assets by design.
Sharpe Ratio	P3 – HRP	HRP achieves the best balance between return and risk across rolling windows.
Sortino Ratio	P3 – HRP	Downside volatility is reduced through diversified cluster allocation.
Max Drawdown	P4 – Denoised+HRP	Denoising stabilises the covariance input while HRP spreads risk evenly, limiting large drawdowns.
CVaR 95%	P3 – HRP	HRP reduces extreme tail losses through diversification across uncorrelated asset clusters.

5. Step 5: Out-of-Sample Test (July 2025) & Final Recommendation

We have the walk-forward backtesting results from Step 4 for all our portfolios. In the walk-forward backtest, P3 (HRP) is the top performer across five of the six metrics. P4 (Denoised+HRP) wins on Maximum Drawdown and, when combining overall robustness with the lowest weight variability (1.71%), is our recommended long-run strategy. In the last iteration of the experiment, the new set of weights were used to re-estimate the covariance matrix from the full training set for the entire training period up to 1 January 2024. The weights were held constant for the entire out-of-sample test period of 1 July 2025 to 31 July 2025.

5.1 Out-of-Sample Performance: July 2025

Table 4 reports Out-of-sample results for the 4 portfolios for the period July 1—31, 2025. The P3 and P4 portfolios gave annualized losses of about -3.8% and their corresponding Sharpe ratios were -1.220 and -1.24 . The current returns for the four portfolios are quite different from each other, but they are still the same four as before. The maximum drawdown in each portfolio so far has been less than 1.6% .

Table 4 – Out-of-Sample Metrics: July 2025 (annualised, $R_f = 5\%$)

Portfolio	Ann. Ret.	Ann. Vol.	Sharpe	Sortino	Max DD	CVaR 95%
P1 – Baseline MV	34.36%	7.21%	4.0712	7.7773	-1.36%	-0.71%
P2 – Denoised MV	31.39%	6.99%	3.7742	6.7556	-1.39%	-0.72%
P3 – HRP	-3.77%	7.08%	-1.2392	-2.3420	-1.48%	-0.73%
P4 – Denoised+HRP	-3.80%	7.20%	-1.2220	-2.3322	-1.50%	-0.75%

5.2 Q1: Why Do the Four Portfolios Perform so Differently OOS?

Metrics for July 2025 out of sample are reported in Table 5 relative to their in sample values. The Sharpe ratios for P1 and P3 have moved by $+2.34$ and -1.96 respectively, which tell two very different stories. about portfolio behaviour under new data.

Table 5 – Performance Change: OOS minus In-Sample (positive $\Delta\text{MaxDD} / \Delta\text{CVaR}$ = smaller loss, i.e. better)

Portfolio	Δ Return	Δ Vol.	Δ Sharpe	Δ Sortino	Δ Max DD	Δ CVaR
P1 – Baseline MV	+6.36%	−6.08%	+2.34	+5.44	+9.64pp	+1.10pp
P2 – Denoised MV	+3.71%	−6.12%	+2.04	+4.40	+9.34pp	+1.06pp
P3 – HRP	−15.51%	−2.22%	−1.96	−3.28	+5.77pp	+0.61pp
P4 – Denoised+HRP	−15.40%	−2.04%	−1.94	−3.26	+5.67pp	+0.58pp

5.3 Q2: Reasons for the Performance Differences

1. Financial product characteristics. The twelve assets span very different economic sectors. In 2025, we saw continued strength in financials and commodity-linked assets. P1 portfolio made a huge bet that those markets would continue to outperform. That bet paid off in July 2025, creating a momentum-like payoff for P1. By contrast, HRP allocated meaningful weight to TLT and UNH, both of which underperformed in July 2025. Because HRP spreads risk widely, it was dragged down by the laggards, even as most of its other stocks rose.

2. Training data and estimation error. The 2024–2025 training window coincided with a broadly bullish equity market, which means the historical return estimates fed into P1’s optimisation overstated the expected outperformance of growth and financial assets. When July 2025 continued that trend, P1 benefited by coincidence. However, the walk-forward backtest – which uses shorter, rolling windows and therefore data that reflects a broader set of market conditions — really does tell a more accurate story: P1’s average annualised return across all rolling windows was -0.65% , well below P3’s 7.62% . This also confirms that P1’s good OOS in July 2025 was not fully reflected in the initial price adjustment calculation. reflects a lucky alignment of market conditions rather than a robust signal (López de Prado, *Building Diversified Portfolios* 3).

3. Mathematical model dynamics. Max-Sharpe optimisation requires inverting the sample covariance matrix. As Michaud (31) showed, this process effectively acts as an error maximiser: the optimiser overweights assets whose estimated returns happen to be too high and underweights assets whose estimated correlations happen to look artificially low. Our sample covariance has a condition number of 36.4, which drops to 35.2 after denoising – both indicate moderate ill-conditioning. A direct empirical test of this instability is the bootstrap weight stability analysis: drawing 50 random subsamples of 80% of the training data and re-estimating portfolio weights each time, P1’s weights vary with an average standard deviation of 5.48% across draws, compared to just 1.69% for P3. This three-fold difference in variability directly quantifies how much more sensitive P1 is to estimation noise.

5.4 Q3: Does the Added Complexity Justify the Improvement?

Table 6 – Incremental Complexity Analysis (OOS, July 2025); positive ΔMaxDD = smaller drawdown

Portfolio	Δ Sharpe vs P1	Δ MaxDD (pp)	Incr. Complexity	Justification
P1 – Baseline MV	0.00	0.00pp	None (reference)	—
P2 – Denoised MV	−0.297	−0.03pp	Low: $\mathcal{O}(N^3)$	Not justified in July 2025
P3 – HRP	−5.310	−0.12pp	Low-Med: $\mathcal{O}(N^2 \log N)$	Context-dependent
P4 – Denoised+HRP	−5.293	−0.14pp	Med: $\mathcal{O}(N^3) + \mathcal{O}(N^2 \log N)$	Justified long-term

The 18-month walk-forward backtest, which covers many different market conditions, gives a much more meaningful picture: P3 and P4 both dominate P1 and P2 across all six metrics in that longer evaluation. Denoising alone (P2) adds very little value at any horizon. HRP-based approaches consistently reduce drawdown, tail risk (CVaR), and weight instability at a low additional computational cost. P4 is therefore the most robust long-run choice, even though July 2025 happened to reward the concentrated momentum bet embedded in P1.

5.5 Final Recommendation

We evaluated the strategies using a composite scorecard that averages the ranks across multiple performance and risk criteria (including Out-of-Sample Sharpe, Sortino, CVaR, Maximum Drawdown, and walk-forward stability). The ranking of the portfolios is shown in Figure 7. The highest ranked portfolio was P4 (The P1 portfolio was the winner of the July 2025) P-sets contest with the highest Out-of-Sample Sharpe ratio. However, as we have already seen, this achievement was due to a short term momentum trading opportunity that just happened to coincide with the short term market trend. Indeed, when applying a walk-forward backtest to the P1 system we can see that the high concentration and large variability of the weights together with their high sensitivity to the returns makes a simple mean-variance optimization very vulnerable to errors in the estimation of the moments as well as to sudden changes in the market conditions. We can also see that the higher weight variability of the P1 portfolio (5.48%) is more than 3 times the one of the P4 portfolio (1.71%). While P1 and P4 may yield a percent or so to each other over the next few weeks in a strongly rising market, over the long run for risk adjusted reliability in a diversified portfolio across a spectrum of volatile and flat market environments, P4 has proven itself to be the better choice for investors.

6. Step 6: Technical Talk – Estimation Error

This section presents a five-part technical discussion on how estimation error affects portfolio construction and how our methodology addresses it.

6.1 Slide 1 – What Is Estimation Error?

The estimation error is the discrepancy between the information derived from a sample and the parameter of the population. In other words, the error in estimating the mean population parameter is called and what is actually true about the underlying distribution of asset returns. If $\hat{\theta}$ is our sample estimate of some true parameter θ , then the estimation error is simply $\hat{\theta} - \theta$. In portfolio optimisation, the two inputs most affected by this problem are the expected return vector μ and the covariance matrix Σ . Both must be estimated from the available data:

$$\hat{\mu} = \frac{1}{T} \sum_{t=1}^T \mathbf{r}_t, \quad \text{Var}(\hat{\mu}) \propto \frac{1}{T}, \quad \hat{\Sigma} = \frac{1}{T-1} \sum_{t=1}^T (\mathbf{r}_t - \hat{\mu})(\mathbf{r}_t - \hat{\mu})^\top \quad (14)$$

The variance of the mean estimator is proportional to $1/T$, meaning that longer data series reduce estimation error in expected returns. For our dataset – $N = 12$ assets and $T = 373$ daily observations, leading to $12 \times 373 = 4466$ individual data points. $q = N/T = 0.032$ – for a large sample, the mean estimator should be fairly accurate. is reasonably stable. However, the deeper problem is how errors in $\hat{\Sigma}$ interact with the matrix inversion step in the Sharpe-ratio optimisation, as explained in the next slide (López de Prado, *Machine Learning* 38).

6.2 Slide 2 – Why Estimation Error Is Especially Damaging in MVO

The Markowitz tangency portfolio requires explicitly inverting the estimated covariance matrix:

$$\mathbf{w}^* = \frac{\hat{\Sigma}^{-1} \hat{\mu}}{\mathbf{1}^\top \hat{\Sigma}^{-1} \hat{\mu}} \quad (15)$$

The degree of this amplification is captured by the condition number:

$$\kappa(\hat{\Sigma}) = \frac{\lambda_{\max}}{\lambda_{\min}} \quad (16)$$

A large condition number means that small changes in the high level input can result in large changes in the low level output. produce very large changes in the inverse and therefore in the portfolio weights. Michaud (31) famously described mean-variance optimisation as an “error maximiser” precisely because of this property: the optimiser systematically pours capital into assets whose estimated returns happen to be inflated by noise, and avoids assets whose correlations happen to look high in the sample.

6.3 Slide 3 – Estimation Error in Our Data: Marchenko–Pastur Analysis

Using Random Matrix Theory one can calculate exactly what fraction of the sample is in the bulk of the spectrum. covariance matrix is signal and how much is noise. Table 7 summarises the key diagnostic quantities.

Table 7 – Estimation Error Diagnostics – Marchenko–Pastur Analysis

Quantity	Value
Number of assets N	12
Training days T	373
Ratio $q = N/T$	0.0322
Noise band lower bound λ^-	0.6734
Noise band upper bound λ^+	1.3909
Eigenvalues inside noise band	4 out of 12
Signal eigenvalues (outside noise band)	8 out of 12
Sample covariance condition number $\kappa(\hat{\Sigma})$	36.4
Denoised covariance condition number $\kappa(\tilde{\Sigma})$	35.2

Four of the twelve eigenvalues of our sample correlation matrix fall inside the noise band $[0.6734, 1.3909]$ and are statistically indistinguishable from those of a purely random matrix. These four eigenvalues carry no reliable information about the true covariance structure and inflate the condition number to 36.4, making the inverse unstable. After denoising the condition number is down to 35.2 so that the noisy inverse does not direct huge amounts of capital to the “most optimal” portfolio based solely on the random sampling noise. (López de Prado, *Machine Learning* 57).

6.4 Slide 4 – How Our Methods Address Estimation Error

Table 8 – Methods and Their Measured Impact on Estimation Error

Method	Mechanism	Measured Effect
Denoising (RMT, P2)	Replace noise eigenvalues in $\hat{\Sigma}$; reconstruct $\tilde{\Sigma} = Q\tilde{\Lambda}Q^\top$	κ for the sector fell by -3.3% to 35.2 from 36.4 and we increased the weight on the TLT to 6.13% from 2.99% to eliminate some spurious short-term covariances
HRP (P3)	Recursive bisection on cluster variance; <i>never</i> computes $\hat{\Sigma}^{-1}$	Error amplification is completely eliminated: - the average bootstrap weight std dev decreased from 5.48% (P1) to 1.69% (P3)
Denoised+HRP (P4)	Denoising cleans $\hat{\Sigma}$ before HRP applies recursive bisection	Lowest weight variability (1.71%) and best composite rank across all 7 criteria

Looking at Table 8 we can see that the largest decrease in error comes from the movements of HRP’s arms. Inversions are not necessary, nor is denoising. Denoising provides a modest, though useful improvement to the performance of the basic reconstruction. Using all the amplification methods together gets you much better input quality and robust method (López de Prado, *Building Diversified Portfolios* 5).

6.5 Slide 5 – Results and Conclusion

Table 9 reports the average cross- bootstrap standard deviation of portfolio weights across the across 50 subsamples, each drawn from 80% of the 373 training days (approximately 298 observations per subsample. Sensitivity analysis to estimation error in a Bayesian model average approach for portfolio optimization (using 80% of training observations) measures the sensitivity of the portfolio performance to the estimation errors of the parameters. Table 9 – Bootstrap Weight Stability (50 iterations, 80% subsamples).

Table 9 – Bootstrap Weight Stability (50 iterations, 80% subsamples)

Portfolio	Avg. Weight Std Dev (Bootstrap)	Stability Rank
P1 – Baseline MV	5.48%	Rank 3 (most sensitive)
P2 – Denoised MV	5.49%	Rank 4 (most sensitive)
P3 – HRP	1.69%	Rank 1 (most stable)
P4 – Denoised+HRP	1.71%	Rank 2 (most stable)

The average bootstrap weight standard deviation for the Sharpe portfolio is 5.48% more than three times as large as the 1.69% for the HRP portfolio. This measures the greater sensitivity of the Sharpe portfolio to the noise in the estimates. Risk to Investor This measures the direct, measurable impact on the investor’s portfolio. In the walk-forward backtest, P1 (Portfolio 1) refers to the investor’s portfolio. has delivered an average annualised

return of -0.65% over rolling windows, while P3 has delivered an average annualised return of $+7.62\%$ over the same period, even when the large weight assigned to future returns is included. Both strategies are run using the same universe of 12 assets and the same historical daily returns.

Denoising and HRP are a cheap way to deal with this issue in practice, at the cost of $\mathcal{O}(N^3) + \mathcal{O}(N^2 \log N)$ computations (López de Prado, *Machine Learning* 71). P4 has the smallest weight variability, the best composite rank and the most stable walk-forward risk. Also, all portfolios experience some degree of in-sample to out-of-sample risk decay as shown in Table 5 but this effect is largely mitigated by the Denoised+HRP pipeline.

7. REFERENCES

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